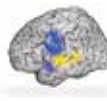




Trump vs Solar Panels

Donald Trump's administration announced that it would be imposing a 30 percent tariff on foreign-made solar panels. <http://dgit.in/TrumpSp>



Researchers track a thought

Neuroscientists have recorded a single thought moving through the brain by measuring the electrical activity of neurons. <http://dgit.in/braintrip>

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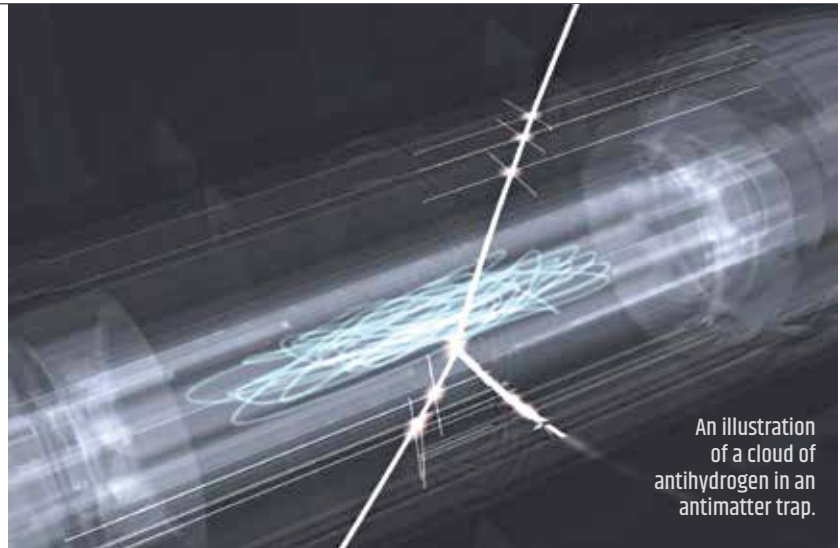
he works of Isaac Newton and Gottfried Wilhelm Leibniz gave rise to the field of classical

mechanics, which described how bodies behaved when various kinds of forces were applied to them. While classical mechanics were good enough to understand the phenomena that most humans experienced on a daily basis, it fell short when investigating the motions of planets, stars and galaxies, or the subatomic realm.

Albert Einstein first published his theory of relativity in 1905, then refined the concept in 1916. A fundamental idea of the theory was that space and time are not separate entities, but instead form a single spacetime continuum.

The investigations of how particles and energies behave in the subatomic realm by researchers such as Max Planck, Niels Bohr, Werner Heisenberg, Wolfgang Pauli, Erwin Schrödinger, Satyendra Nath Bose, and Albert Einstein gave rise to the field of quantum mechanics. Quantum here, just means extremely small spaces. Special relativity had not been described in the context of quantum mechanics till Paul Dirac came along, leading to a weird and wonderful implication.

Within an atom, the proton has a positive charge, while the electron has a negative charge. A shortcoming of quantum mechanics at that time was that it could not explain the motions of subatomic particles at high speeds. Dirac wanted to explain the motion of electrons travelling at speeds close to the speed of light. The equation he came up with to do so, was sufficient enough for his purpose, but had far reaching consequences. Dirac's equation was consistent with both the theory of relativity as well as quantum mechanics, and is in some ways similar to the Schrödinger equation. Dirac's equation had an inherent "problem". Just as quadratic equations have two solutions, or roots, Dirac's equation also had two solutions. The equation implied that



An illustration of a cloud of antihydrogen in an antimatter trap.

Image: CERN/Chukman 50

WHY IS ANTIMATTER SO RARE?

Antimatter incredibly expensive to create, very hard to store, and cannot come in contact with regular matter without disappearing in a flash of gamma rays and neutrinos.

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the electron had an antiparticle, with a positive charge instead of a negative charge, now known as a positron.

Carl David Anderson, while studying cosmic rays in a magnetic environment, found traces of a particle that was moving in the opposite direction of the electron, but was too light to be a proton. The new particle was called the positron, and Anderson won the 1936 Nobel Prize in Physics for the discovery. After the discovery, the results of previous studies showed that the positron had been experimentally observed before, but such observations were not followed up, or the particle was assumed to be a proton.

PARTICLE ACCELERATORS

The idea that every particle had a counterpart with an opposite charge was known as charge symmetry. The Bevatron was a device designed to specifically test the validity of charge symmetry. It was a particle accelerator, that would use electromagnets to propel protons to speeds close to that of light, and then make them collide against stationary protons or neutrons. In 1954, the Bevatron became operational at the Berkeley Lab, operated by the University of California.

Just about a year later, the first antiprotons were observed at the Bevatron. A year after that, the first